

GRIDBENCH · DESIGN DECISIONS REGISTER

GridBench design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/gridbench		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-01	Amish Chadha	Register opened with the open decisions, items to confirm, value engineering and decisions made to date

GridBench design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md` , GBN-BLD-001) describes the design as it stands and does not list open decisions.

■ Open decisions

Table 1. Open decisions.

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	Review of the design-for-construction changes (bolt and cross dowel frame joints, notched rail, shorter shelf, worktop brackets, tile pocket and fixing, T-nuts, fixtures on threaded positions, arm clamps, wall anchor)	(a) accept as recorded; (b) amend named changes	(a): none changes what the bench does, its pitch or its safety case	The whole build plan follows them	GBN-DDR-003
2	Clamp rating at the 122 screw-in positions over the frame, where R7 is not met (pull-out factor 0.57 to 0.92)	(a) rate them light duty at 0.92 N·m (296 to 493 N at the part) and mark them on the worktop; (b) move the insert sub-grid off the frame lines; (c) relax R7 at those positions	(a)	How the worktop is marked; safety stop S5	GBN-DDR-002, O4; GBN-CAL-001 [G5e]
3	Racking stiffness of the frame	(a) end low rails between the legs at each end; (b) screw the shelf down to the low rails; (c) leave as is until a TRL 4 push test	None yet	Steps 4 and 7	GBN-DDR-002, O3
4	First workshops and regions for co-design	Workshops and regions to approach first	None yet	Not part of the TRL 3 build	GBN-DDR-001, O1
5	Wall anchor angles in the product renders	(a) leave them out of the renders and keep them in the model, drawing and BOM; (b) render them with a wall	(a)	None (renders only)	docs/REVIEW.md , 2026-09-26, item 2
6	Full hole pattern in the renders, representative patch in the exported CAD	(a) keep the split; (b) export every hole	(a): the full pattern makes very large STEP and STL files	None (renders and exports only)	docs/REVIEW.md , 2026-09-26, item 3
7	Appearance-only details in the renders (index ticks, name plate, bolt heads, knobs, sample block and bar)	(a) render detail only; (b) adopt them into the design and the BOM	(a)	None unless adopted	docs/REVIEW.md , 2026-09-26, item 4

The fourth appearance item of 2026-09-26, the toe clamp screws on tile holes, is settled by GBN-DDR-003: the model now puts them there too.

■ To confirm when parts are bought

Table 2. Items to confirm.

#	What to confirm	Why it matters	Source
1	The tile offcut's real thickness, flatness and price	The packers are planed to suit its thickness; R5 depends on its flatness; the USD 28 price is unverified	GBN-CAL-001; BOM line 6
2	The plywood's real thickness	Packer thickness and the 15 to 18 mm screw reach below the surface both assume 18	GBN-DDR-003, P6 and P9
3	The tee nut's flange (19), barrel (9.5) and hole size (8.0), and the screw-in insert's hole size (8.5)	The worktop is drilled to these sizes before the parts can be tried	BOM line 5
4	The M8 cross dowel's diameter (10) and the bolt length (120)	They set the cross dowel holes and how far the bolt reaches into the member	GBN-DDR-003, P1
5	The levelling foot thread and the T-nut's barrel size	They set the 12 mm bore in the leg end	GBN-DDR-003, P8
6	The 20 x 40 extrusion's core hole size	The post is held by M5 screws tapped into its core holes	GBN-DDR-003, P10
7	A sourced figure for plywood in-plane moisture movement	R3 uses an unsourced 0.007 % per % moisture content	GBN-CAL-001 [E2]
8	Whether the tile can be drilled to ± 0.05 without CNC, with a drilling jig on a drill press	R4 hole position cannot be verified on paper	GBN-PRC-001, Open questions

■ Value engineering

Value-engineering target: USD 250 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 268.40 (USD 18.40 over the target) for the core parts, from `bom/bom.csv` [K3], [K4]; with the user-supplied dial indicator and the optional wall anchor, USD 288.40. Main cost drivers and savings worth trying:

- **Frame, USD 57.00.** About USD 33.60 of timber and USD 23.40 of bolts, cross dowels and screws [K2]. Worth trying: one bolt instead of two at each low rail end (about USD 3.20); bolts and cross dowels bought as a bulk furniture-fitting pack; or a glued and screwed frame (about USD 15 less, but it no longer comes apart).
- **Worktop, USD 45.00.** A half sheet of birch plywood plus router wear. Worth trying: a full sheet shared with the shelf, or a cheaper sheet faced only on the top.
- **Threaded inserts, USD 28.20.** Worth trying: a bulk price for tee nuts and inserts; leaving the 122 screw-in positions as plain holes if decision 2 rates them light duty anyway.
- **Precision tile, USD 28.00.** An unverified offcut price. Worth trying: offcut dealers' minimum-cut prices; a 250 x 250 tile if the fixture library allows.
- **Instrument post and arm, USD 15.00, and fastener kit, USD 14.00.** Worth trying: steel bar and screws from bulk stock.

■ Decisions made

Table 3. Decisions made.

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items A1 to A10: cost options (dial indicator user-supplied, tile as an offcut), 25 mm M6 grid, two zones, 50 mm insert sub-grid, bolted timber frame, printed PETG fixtures, ballast plus optional wall anchor, fixture metadata block, CERN-OHL-S-2.0 fixture library, tile at the right	Amish: "i accept all your recommendations, go with them across all repos."	GBN-DDR-001, GBN-DDR-002
2026-09-25	budget_usd raised from USD 220 to USD 250 (O2)	Amish, same instruction	GBN-DDR-002
2026-09-25	Flanged tee nuts from the underside, screw-in inserts only over the frame (D1); 45 x 120 mm aprons (D2); rubber-padded feet (D3)	Amish, same instruction	GBN-DDR-002
2026-09-30	Build plan format approved; outstanding decisions kept in this register, not the build plan; the design to be made physically buildable as the pictures are drawn	Amish: "this is the correct build plan ... this is a good quality document format. Extend this across all the other repos"; "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."	GBN-BLD-001, GBN-DDR-003
2026-10-01	budget_usd is a hypothetical value-engineering target, not a limit; cost is reported as over or under it	Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens. its ok to ensure wording reflects that the hypothesis budget was x - the real cost being accrued is y"	This register, Value engineering